

Title: Mini Shell and Tube Heat Exchanger

VILLANOVA UNIVERSITY Dept. of Mechanical Engineering Course/Section: ME 3333-002

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Objectives

The objective of this report was to outline material requirements and produce a preliminary quantitative parametric analysis to determine target design points for each component. Additionally, the ideal manufacturing process was outlined to construct the product by combining each part. Later, the true intended manufacturing process was discussed, and detailed drawings of the product were produced. Finally, the prototype was manufactured and evaluated.

2 Project Design

Proposed Project

The proposed project is a mini heat exchanger (HX), about a cubic foot in volume, with the intended goal of acting as a miniature air control product. The HX should be capable of heating air to different intended temperatures within its operating capability. The product would use piping to hold hot water for heat transfer to occur, an outer shell to trap air within the system and direct it out to a single point, a base to provide structural support for the product, a motorized fan to blow ambient air through the system, and panels to hold pipes in place securely.

Part Requirements and Proposed Materials

Copper piping was used due to its high thermal conductivity (413 W/mK) and high heat resistance. Copper piping is also corrosion and rust-resistant, allowing the pipes to handle water effectively without worrying about deterioration. The high melting point of copper (1084 °C) will sufficiently meet the heat resistance requirement.²

Aluminum outer caps were selected due to their relatively high thermal conductivity (237 W/mK) and light weight. Compared to steel, aluminum has a significantly higher thermal conductivity as well as being a lighter alternative. Since this will be a desktop space heater, it must be sturdy enough to withstand heat, while also being light enough to maneuver easily.²

Polystyrene was selected as the outer shell material due to its low thermal conductivity, ability to withstand high temperatures, and light weight. The outer shell's purpose is to prevent heat loss in the system (act as an insulator that funnels air). Furthermore, the copper tubing will be hot, and polystyrene can withstand temperatures around 200°C. The system is designed to sit on a desk, so the polystyrene's light weight will make it easy to move around.¹

The base must be strong enough to withstand the weight of all other system components (the shell, copper tubing, fan, outer caps, and hot fluid in the pipes). We have selected stainless steel for its material strength of 200 MPa. Over our product's projected footprint of 10" by 20", our plate would be able to withstand over 2,347 kg of weight force (well beyond the compressive strength and weight we plan expose the plate to). The base must be in contact with the fluid water, so it must be non-rusting.⁵

To better understand the thermal capabilities of the proposed system, a sample calculation was performed by modeling the system as a crossflow heat exchanger where the hot water was unmixed, and the cold air was mixed. Assumptions were as follows:

$$\dot{m}_h = 0.02 \frac{kg}{s}, c_{ph} = \frac{4184J}{kgC}, \dot{m}_c = 0.001 \frac{kg}{s}, c_{pc} = \frac{1000J}{kgC}$$

Modeling the system as a thermal circuit yielded an NTU value of 0.16, correlated to an effectiveness of about 10%. These calculations made sense considering the small scale of the system and likely inefficient heat transfer. Using the following equation:

$$\varepsilon = \frac{q_{hx}}{C_{min}\Delta T_{max}}$$

The power output of the system was determined to be 7W, which yields a temperature increase of the incoming cold air by 7°C before it is blown out the exit vent.

The mass flow rate values assumed above were applied to determine the required property values for the fan and motor system, as tabulated data can project a mass flow rate requirement in terms of required revolution speed based on the area that the fan is covering.

To justify the material choice for the outer shell, a cylindrical thermal circuit model can be applied to the system using the following equation for the conductive resistance of the polystyrene:

$$R = \frac{\ln(\frac{D_o}{D_i})}{2\pi kL}$$

Since the power output was previously determined to be 7W, and the thermal resistance can be calculated to be approximately 2.5 K/W, the temperature difference between the inner and outer walls of the system will return to room temperature due to the influence of exterior convection. This makes the shell safe to the touch. The maximum thermal conductivity was calculated to be 0.2 W/mK. The other insulation consideration was off-gassing (toxicity when the insulation heats up). Research shows that many ethylenes and some propylenes meet this requirement.¹ To minimize the thermal conductivity while maintaining strength in the foam, expanded polypropylene was selected for its high material strength compared to other foams (0.8 MPA) and its low thermal conductivity (0.032 W/mK).¹

A complete layout of all required design parameters and materials best suited to fulfill all design requirements is seen in Table 1.

Table 1. Comparison of Part Requirement and Material Properties

Part Name	Name of Required Property	Value of Required Property	Proposed Material	Material Property	Reference (put # in)
Piping	Conductivity (W/mK)	$k > 200$	Copper	413 W/mK	2 3
	Melting Point (°C)	$MP > 200^{\circ}\text{C}$	Copper	1084°C	
Outer Caps	Conductivity (W/mK)	$k > 200$	Aluminum	237 W/mK	1 3
Shell	Conductivity (W/mK)	$k < .2$	Polypropylene	0.032 W/mK	1 3
	Melting Point (°C)	$T > 150$	Polypropylene	160°C	
Fan and Motor System	Revolution Speed	$\omega > 100 \text{ RPM}$	PLA Plastic (Fan), Stock Metal DC Motor	1000 RPM (Motor)	4
	Mass Flow Rate Capacity	$\dot{m} > 0.001 \text{ kg/s}$		6'' x 6'' Area	

¹ “Looking for Non-Off Gassing Foam? These Are Your Options - Amcon Foam.”

² “Thermal Conductivity of Metals and Alloys: Data Table & Reference Guide.”

³ “Metals and Alloys - Melting Temperatures.”

⁴ “The 3 Fan Laws and Fan Curve Charts.”

Base	Compressive Material Strength	P>20	Stainless Steel	200 MPa	5
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Proposed Manufacturing Technologies

This project will combine multiple types of manufacturing. The material justification for the fan's propeller dictates PLA, so additive 3-D printing technology will be used to print the blade. This is the optimal technology for ensuring the correct shape of the blades is manufactured, which would be difficult on a subtractive manufacturing machine due to the precise curves of the blade.

The copper piping will be bought in stock but cut to the precise lengths using a hand-held pipe cutter tool, then the ends will be surfaced with a belt sander to ensure equal lengths and flat surfaces on both ends for optimal construction.

The outer caps will be the holding pieces for all of the copper piping, so the holes will be measured and milled to ensure equal diameters to fit flush with the copper piping. Ideally, these would be secured by welding the pipes to the plates, but this small-application prototype super-glue will suffice.

The base will be cut to size using subtractive manufacturing, and although it would ideally be a heating plate to maintain constant water temperature through the pipes, a metal sheet to seal the copper tubes will suffice for this prototype. It will have to be cut on a band saw or surfaced with a mill to acquire the desired dimensions.

The outer shell will be made of foam insulation, and since it will need to be cut to specific shapes to surround the hexagonal pattern of the end plate, it will be cut to precise angles and lengths using the desktop CNCs in Drosdick 012, which we have been learning to use in ME3900. The base will have rubber feet cut using subtractive manufacturing.

The base will be cut from a sheet of stainless steel. We will recycle waste sheets of metal and cut our desired rough area. Then we will round the edges and polish the surface on the mill until safe to handle and prepped for welding. At a later stage, the base will be TIG-welded to the copper pipe.

3. Results and Discussions

3.1 Finalized Design

⁵ "The Joining of Copper to Stainless Steel by Solid-State Welding Processes: A Review."

The final design of the heat exchanger consists of six parts: the copper pipes, the aluminum caps, the steel base, the foam insulation walls, the 3D-printed aluminum fan, and the 12V DC motor to control the fan. All parts will be manufactured in-house except for the DC motor. Figure 1-6 include 3D exploded models and drawings of the assembled product, as well as each of the individual parts that will be manufactured.

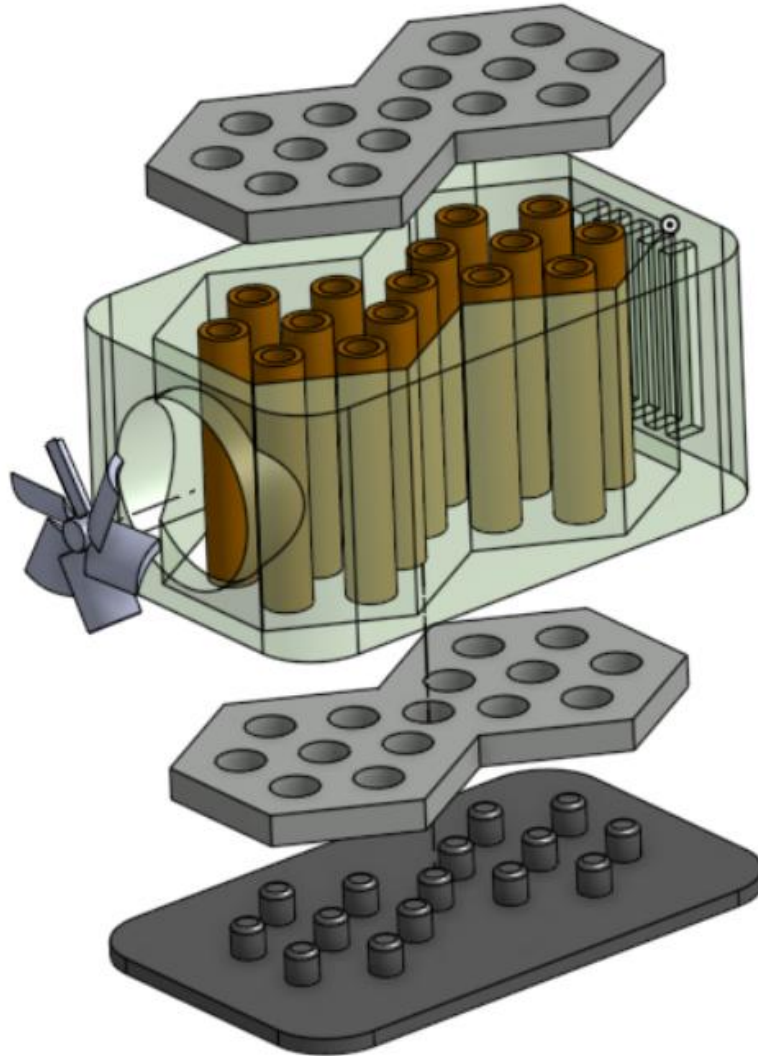


Figure 1. Exploded Assembly of Heat Exchanger Designed in SolidWorks

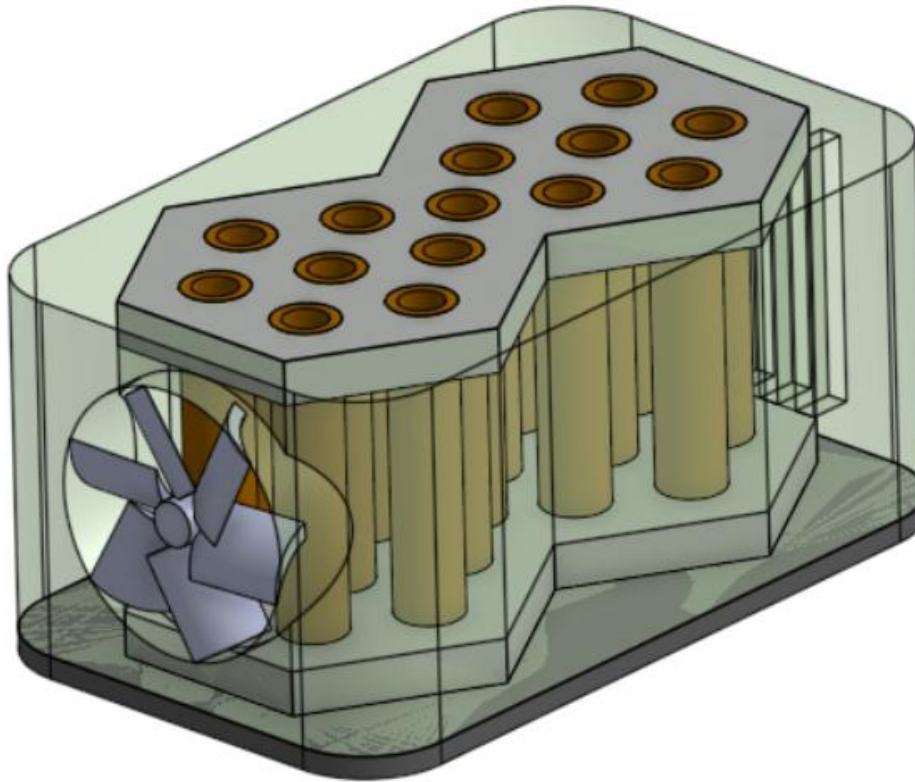


Figure 2. Unexploded Assembly of Heat Exchanger Designed in SolidWorks

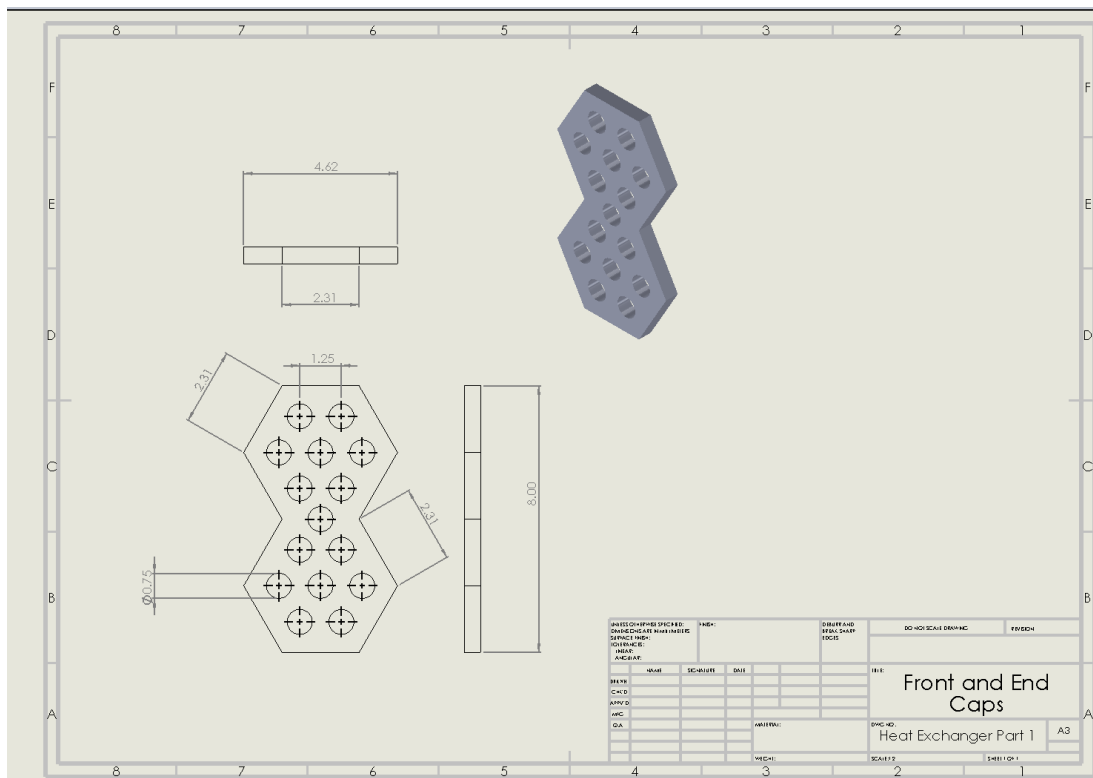


Figure 3. Engineering Drawing of the Aluminum Caps

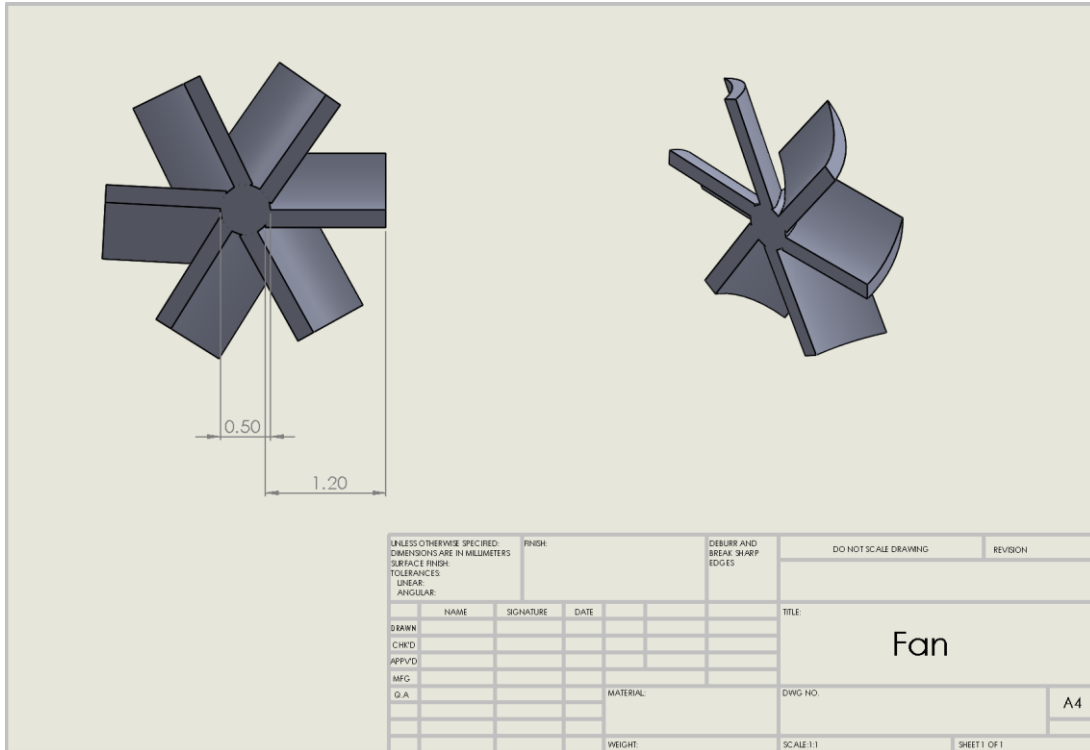


Figure 4. Engineering Drawing of the 3D Printed PLA Fan

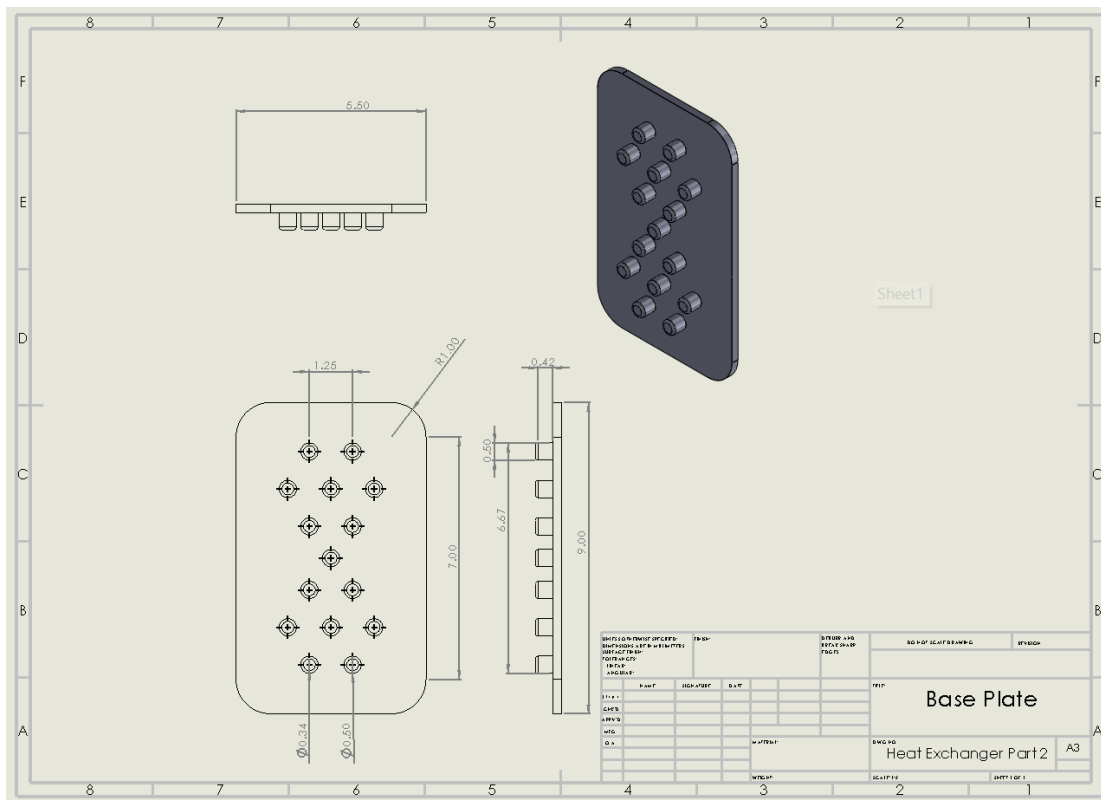


Figure 5. Engineering Drawing of the Steel Base

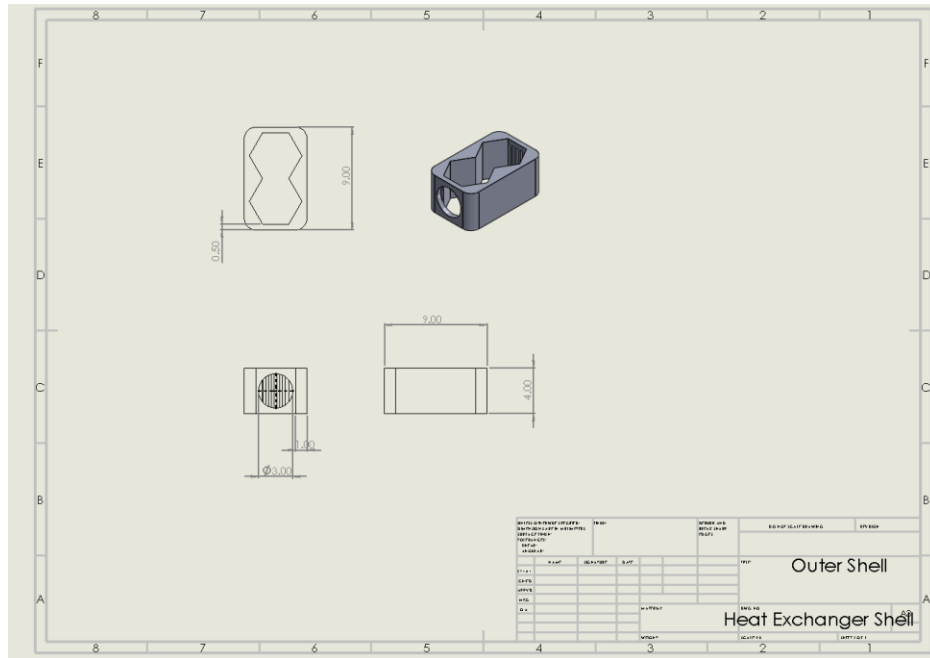


Figure 6. Engineering Drawing of the Assembled Foam Insulation Walls

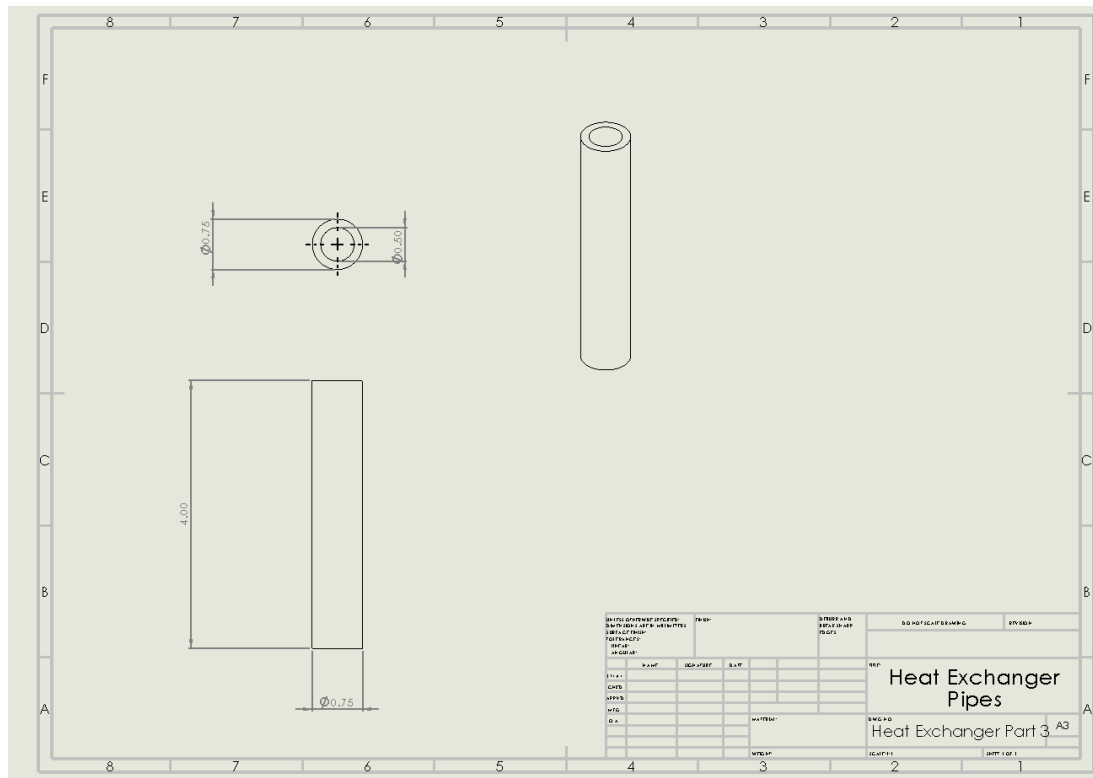


Figure 7. Engineering Drawing of the Copper Pipe

The final product will consist of six main components. The copper pipes, the aluminum caps, the insulating walls, the base, and the fan will be manufactured in-house, so materials need

to be acquired for those, seen in Table 2. The sixth part is the motor to control the fan, and that will be purchased.

Table 2. Material List of the Manufactured Products

Material	Quantity	Unit Price (\$)	Overall Price (\$)	Potential Vendor
Cu – Copper Tubing	15	5	75	Recycled Scrap Machine Shop
C ₂₇ H ₃₆ N ₂ O ₁₀ - polyurethane (foam)	1 (multiple pieces)	2	2	ME Lab Inventory
AISI 4140 Alloy Steel	1	15	15	Recycled Scrap Machine Shop
Al-Aluminum Cap	2	15	30	Machine Shop
C ₃ H ₄ O ₂ - PLA Fan	1	0.89	0.89	Innovation Lab Inventory
12 V DC Motor	1	12	12	Amazon DC Motor

3.2 Finalized Manufacturing Technologies

The copper pipe is received in long sections, so the first order of manufacturing is to use pipe cutters to cut our pipe to the desired length, under $\pm 1/8$ inch of tolerance. Then on the grinding wheel at a speed around 3000 RPM, we will smooth the ends of the pipe to an Ra (average roughness) ≤ 0.1 mm. The inner and outer surfaces need a wash of water. A water jet of 50 m/s can effectively loosen dust and dirt collected on the faces of our pipes. After drying, the application of mineral oil to the outer faces will reject water corrosion and increase longevity. Each section of copper piping will be joined with adhesives to the aluminum caps, which are connected to the whole body of our product.

The aluminum caps for the heat exchanger will be cut to shape using subtractive manufacturing. The shape of the part will be cut on the band saw. For aluminum alloys, the optimal blade speed is around 300 FPM.⁶ There will also be small indents in the caps where the copper pipes will fit into to allow for more seamless fastening. The indents will be milled at an

⁶ “A Brief Overview of the Speeds on Your Band Saw.”

optimal range of 4000-7000 RPM.⁷ As listed in the previous section, the aluminum caps will be connected to the copper pipes with an adhesive application, fastened together to create a stiff, static connection plane.

The fan will be developed with additive manufacturing following a design that was produced with SolidWorks. The print will utilize a layer height of 0.2 mm, 2 wall loops, a 50% infill density with a simple grid pattern, and the support feature will be enabled due to the complex geometry of the fan blades. The SolidWorks model will be sliced in Bambu Studio and printed on a Bambu Lab X1 Carbon printer with a 0.4 mm diameter nozzle with Bambu PLA plastic that reaches approximately 200-220 °C during printing. This printer can be found in the Drosdick Innovation Laboratory. Print time is approximately an hour, and the completed print will use around 30 grams of filament, with the device being around 25 grams after printing.

The base of the product will be cut out of a 1/2" aluminum panel. It will be cut to shape on the band saw, at an optimal rate of around 300 FPM. It will be surfaced on the mill to ensure a smooth finish and optimal dimensions to fit seamlessly with the walls and copper piping. The piece will be milled with a 1/2" diameter cutting tool.

The insulation walls of the heat exchanger will be manufactured in multiple pieces. To successfully fit the hexagonal shape of the aluminum caps, the pieces will be manufactured on the desktop CNCs in Drosdick 012. The cutting tool should be 1/8" in diameter to maintain the necessary precision. The 3D models of the walls will be uploaded to the CNCs and cut, then fastened together using adhesive to complete the entire surrounding wall.

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3.3 Evaluation of Prototype

The constructed prototype can be seen in Figure 8 with all attached components besides the battery pack for the fan and motor system. The top portion of the foam insulator was removed to view the interior aluminum caps and copper piping.

⁷ "Milling Speed for Aluminum: Optimal Rates & Tips."

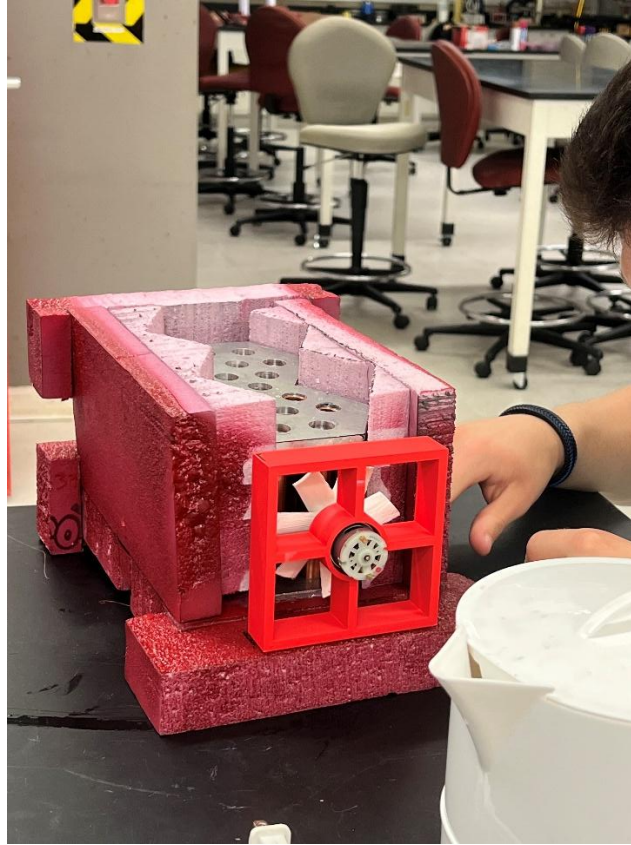


Figure 8: Prototype

The prototype was proposed to increase the temperature of the air that exited the system compared to the ambient air that was taken in by the fan and motor system. This prototype intended to serve as a battery-powered, desktop space-heater. Preliminary calculations indicated that the device would have the capacity to heat the air by 7 degrees Celsius. The prototype came close to fulfilling this function given its ability to heat the air by 5 degrees Celsius.

The main “heat exchanging” components of the system, the fan and motor as well as aluminum and copper piping arrays, were able to be constructed as intended. The ½” copper piping were fastened into the aluminum caps that were constructed through subtractive manufacturing milling. However, the foam insulation layer was too large to be constructed with the CNC machines that were available to the team. As a result, the vertical bandsaw was utilized to cut the key components of the foam insulation layer, which was then glued together. This compromise did not hinder the function of the prototype in any way. However, this compromise may have resulted in a less aesthetically pleasing product. The team was still able to get a proper seal to the piping array and ensure that there was no water leaking from the system.

Conclusions

The objective of this report was to design a mini heat exchanger and perform parametric analysis to find adequate design points. The final design used two, half inch aluminum pieces for the end caps, 15 half inch copper pipes, polyurethane foam for the outer shell, and a 3D printed fan that was connected to the motor. These were chosen due to the initial parametric analysis verifying the materials that were chosen. Furthermore, upon testing, the results found in the original parametric analysis were able to hold true. One recommendation that could be implemented was to improve the design of the end caps. A more accurate system can be implemented to make sure the holes on the end caps are lined up correctly. Improperly lining up the end caps will not allow for the copper piping to line up properly.

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